MIL-T-13659 (Ord)
21 September 1954
SUPERSEDING (In Part)
Army 51-72-34A
26 July 1945

# MILITARY SPECIFICATION

# TELESCOPE, ELBOW, M24A1

## 1. SCOPE

1.1 This specification covers an elbow telescope used with Telescope Mounts M28 and M31 in direct azimuth sighting of 90-MM and 120-MM Gun Mounts.

## 2. APPLICABLE DOCUMENTS

2.1 The following specifications and drawings of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-P-12101 - Packaging of Telescope, Elbow, M1A1, M2, M2A1, M16 Series, M24A1, M26A1, and M62 Series.

U. S. Army

51-7 - Instruments, Fire-Control; General Specification Governing the Manufacture and Inspection Of.

51-70-2 - Optical Components for Fire-Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection Of.

DRAWINGS

Ordnance Corps

91-108-1 - Telescope, Elbow, M24A1

F7680673 - Fixture, Pressure Testing

(Copies of specifications, standards, drawings, and publications required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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## 3. REQUIREMENTS

3.1 General.- Unless otherwise specified, the requirements common to all fire-control instruments as specified in Specifications 51-7 and 51-70-2 shall be mandatory.

3.2 Material.- Material shall be as specified on the applicable drawings and specifications forming a part of this specification.

3.3 Eyepiece focus.- The reticle shall be in sharp focus at the center of the field when the eyepiece is set between minus 0.75 and minus 1.0 diopter. A calibrated dioptometer of at least 3 power, or an equivalent auxiliary telescope, shall be used to make this setting.

3.4 Collimation.- The optical axis shall coincide with the geometric axis, as established by the 1.375 inch diameter and the locating pad, within 0.5 mil.

3.5 Plumb of reticle.- With the axis of the 1.375 inch diameter mounting surface horizontal and the face of the locating pad in a vertical plane, an imaginary line connecting the tops of the range marks shall not deviate from the image of a horizontal target line by more than the angle subtended by a 0.4 mil. block at one end of the reticle pattern when the other end is in coincidence with a level line.

3.6 Image tilt.- The image of a plumb line shall be within one degree of the vertical as measured by an auxiliary telescope in back of the eyepiece.

3.7 Reticle accuracy.- The accuracy of any reticle graduation shall be within 2 percent of the indicated value in mils plus 0.5 mil.

3.8 Resolution.- The resolving power on the optical axis for a parallel-bar test target shall be 10 seconds of arc or better.

3.9 Parallax.- With the target at 900 yards plus or minus 100 yards, the parallax at the center of the field shall not exceed 0.5 mils (approximately 1/2 reticle width).

3.10 Vibration.- After being subjected to the vibration test specified in Specification 51-70-2, except that 2 to 2-1/2 minutes shall be substituted for 15 to 18 seconds, there shall be no damage to the instrument, no change in collimation in excess of 0.2 mil, and no loose dirt in excess of the following:

3.10.1 Dirt on the adjacent surfaces of the objective and the amici prism shall not exceed a total of three in number of a size not larger than a 40 dig.

3.10.2 Dirt on the adjacent surfaces of the prism and the reticle shall not exceed a total of three in number of a size not larger than a 2 dig.

3.10.3 Dirt on the adjacent surfaces of the reticle and the field lens shall not exceed a total of four in number of a size not larger than a 2 dig.

3.10.4 Dirt on the adjacent surfaces of the field lens and eyelens shall not exceed a total of four in number of a size not larger than a 20 dig.

3.11 Sealing.- The telescope shall show no evidence of leakage when subjected to the pressure test specified in 4.3.

3.12 Temperature.- The telescope shall meet the temperature requirements for type II instruments as specified in Specification 51-7.

3.13 Illumination.- The reticle markings shall appear clearly defined when observed in a dark room utilizing the instrument light specified for the telescope.



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3.14 Workmanship.- Workmanship shall be as specified in Specifications 51-7 and 51-70-2.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.- Unless otherwise specified, each telescope shall be inspected for conformance with this specification and Specifications 51-7 and 51-70-2.

4.2 Definition test.- The test object shall be of the pattern illustrated in Specification 51-70-2, figure VI-A. The image shall be viewed with the aid of a 6-power auxiliary telescope of approved design, scaled in diopters. The image of the test pattern shall show correct line count, and the pattern shall be resolved in each of the four meridians by an adjustment not to exceed plus or minus 1/8 diopter from the setting for the best focus of the numeral in the center of the test field.

4.3 Sealing test.- The sealing test shall be performed as specified on Drawing F7680673.

4.4 Temperature test.- The temperature test shall be performed as specified in Specification 51-7.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be as specified in Specification MIL-P-12101.

6. NOTES

6.1 Ordering data.- Contracts and invitation for bids should specify the title, number, date of this specification, and the type of packing.

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